

Student Resume Document
(PDF / DOCX Format)

Target Job Description (JD)
(Role Requirements & Hard Skills)

Stage 1: Spatial Layout & Bounding Box Extraction

PyMuPDF & Tesseract OCR
(Text Stream + Bounding
Boxes [x0, y0, x1, y1])

LayoutLMv3 Multimodal
Tokenizer
- 2D Positional Embeddings
- Visual Patch Encodings
- Text Token Sequence

Structured Document
Entities
- Header / Contact Block
- Education Hierarchy
- Work Experience &
Projects
- Technical Skills Sub-
blocks

Stage 2: Dense Semantic Encoding (Sentence-BERT)

Sentence-BERT Model ('all-
mpnet-base-v2')
(Mean-Pooled Dense
Representation)

Job Description Semantic Vector $\vec{h}_{jd} \in \mathbb{R}^{768}$

Resume Semantic Vector $\vec{h}_{res} \in \mathbb{R}^{768}$

Stage 3: Quantified Impact & Rule-Based Heuristics

Spatial Formatting
Penalizer
(Detects unreadable
tables, multi-column
clashes)

Curated Technical
Taxonomy Matcher
(ESCO / O*NET CS
Concepts & Frameworks)

Regex & Spacy Numeric
Matcher
(Detects %, \$, ms, scale
metrics in bullet points)

Stage 4: Multi-Factor ATS Scoring Synthesis

Hard Skill Exact Match Recall $\vec{h}_{skills} = |\vec{h}_{res} \cap \vec{h}_{jd}| / |\vec{h}_{jd}|$

Quantified Bullet Points Ratio $\vec{h}_{ratio} = \vec{h}_{quant} / \vec{h}_{total}$

Dense Semantic Similarity $\vec{h}_{dense} = \cos(\vec{h}_{res}, \vec{h}_{jd})$

Multi-Criteria Synthesis Function $\vec{h}_{ATS} = \vec{h}_{dense} + \vec{h}_{skills} - \vec{h}_{format}$

Stage 5: Output Artifacts & SPV Mutations

**Actionable Student
Remediation Report**
- Missing Hard Keywords
(e.g., Kafka, Docker)
- Weak Action Verb Flags
(e.g., 'Helped', 'Worked on')
- Unquantified Experience
Bullets Identified
- Layout Parser
Degradation Warnings

**SPV Invariant Feature Updates
(M01)**
- F11: ats_resume_score [0.0,
100.0]
- F12:
resume_quantified_bullets_ratio
[0.0, 1.0]
- F13:
technical_skills_match_ratio
[0.0, 1.0]